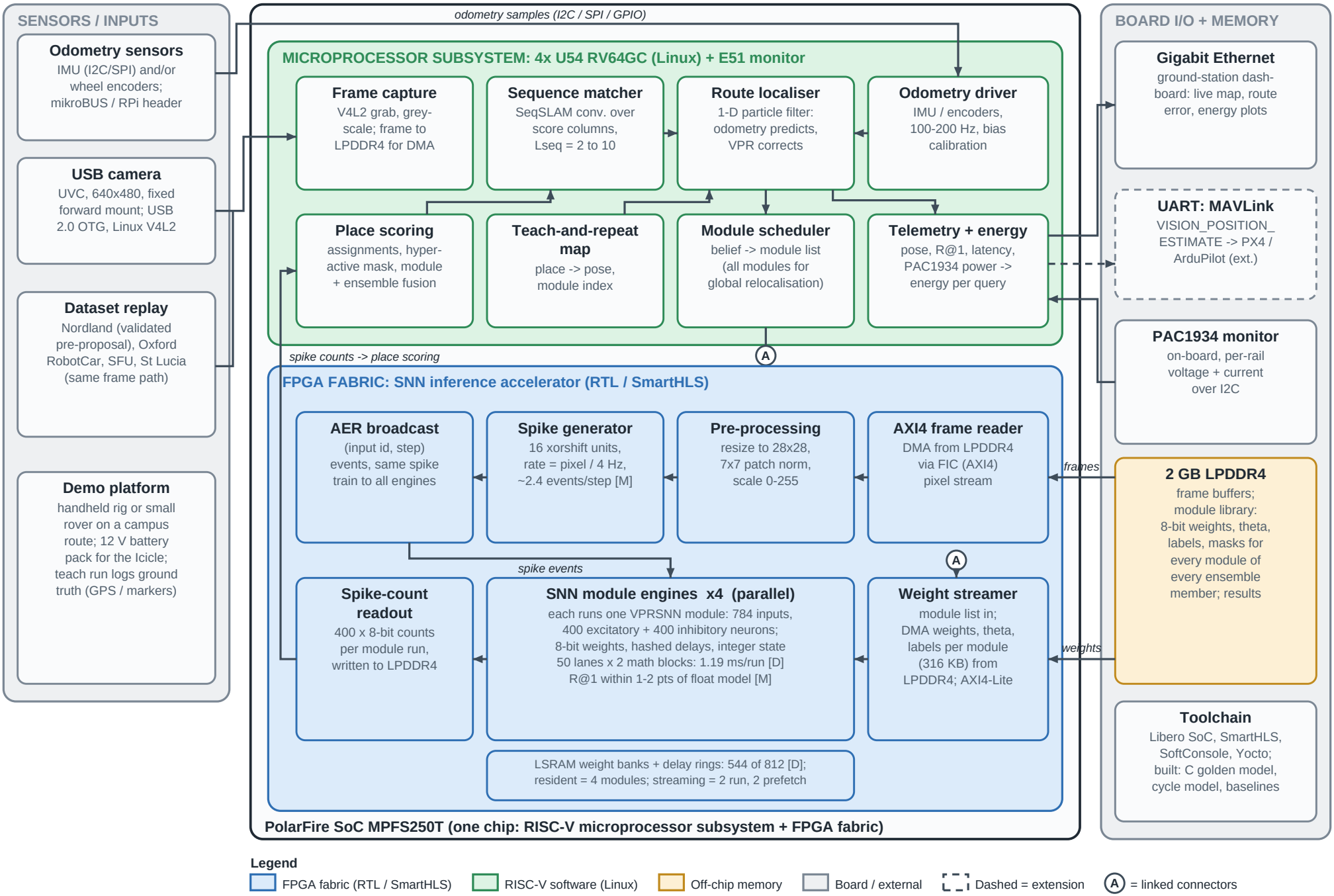
SpikeNav: Real-time Spiking Place Recognition for GPS-denied Robots

System Block Diagram | Track 2: PolarFire SoC Icicle Kit (MPFS250T) | Team: Merconez



Pre-proposal Evidence: Measured Before Writing a Line of RTL

Run by Merconez on a DGX Spark (GB10), reproduced bit-exactly on a Jetson Orin Nano; authors' MIT-licensed VPRSNN code + Nordland

VERIFIED FIDELITY CHAIN (solid = done before the proposal, dashed = planned)

Brian2 reference
VPRSNN code, 3 seeds; R@1 0.954 [M]

→

Float model
spike-exact vs Brian2: 728 / 728 spikes [M]

→

Fixed-point model
integer spec; 12 approximations tested

→

C golden model
32,000 / 32,000 counts; same on Jetson CPU+GPU [M]

→

RTL testbench
vs C golden model, spike by spike

→

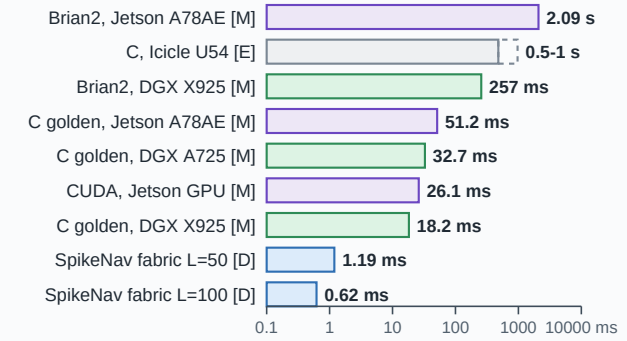
Icicle board
target: identical counts to C model

FULL-SCALE ACCURACY: Nordland, 2,700 test places, authors' released weights [M]

Configuration (Poisson seeds P0 / P1)	R@1	R@5	SeqSLAM L4	SeqSLAM L10
float model (Brian2-exact)	0.549 / 0.553	0.692 / 0.688	0.886 / 0.872	0.976 / 0.964
SpikeNav hardware configuration	0.544 / 0.543	0.692 / 0.691	0.856 / 0.861	0.966 / 0.969
+ masks from our own engine	0.532	0.684	0.860	0.967
no synaptic delays	0.448	0.667	0.821	0.957
hardware configuration, dt = 0.5 ms	0.338	0.594	0.753	0.928

Hardware configuration = 8-bit weights, hashed 1 ms delays, hard reset, shared xorshift spike train, integer state. Cost vs float: about 1-2 R@1 points; with sequence matching (L10) within 1 point.

TIME PER MODULE RUN, ALL PLATFORMS (log scale)



HARDWARE-APPROXIMATION ABLATION: 100 places, 3 model x 3 Poisson seeds [M]

Configuration	R@1	dR@1 vs Brian2	Ensemble-3 R@1	Result
Brian2 reference	0.954 ± 0.007	-	0.998	ref
(a) float model	0.963 ± 0.003	+0.9	0.993	pass
(b) 8-bit weights	0.947 ± 0.015	-0.8	0.997	pass
(c') 4-bit weights, per-module scale	0.958 ± 0.010	+0.4	0.993	pass
(d) hashed delays, 0.1 ms	0.957 ± 0.009	+0.3	1.000	pass
(e') hashed delays, 1 ms binned	0.963 ± 0.015	+0.9	0.993	pass
(f) no delays	0.944 ± 0.005	-1.0	0.997	pass *
(g) dt = 0.2 ms	0.960 ± 0.007	+0.6	0.997	pass
(g) dt = 0.5 ms	0.884 ± 0.017	-7.0	0.987	FAIL
(h) hard reset per module run	0.952 ± 0.007	-0.2	0.990	pass
(i) one spike train shared by modules	0.947 ± 0.012	-0.8	0.997	pass
(j) xorshift spike generator	0.960 ± 0.012	+0.6	0.997	pass
(k) integer (fixed-point) state	0.962 ± 0.017	+0.8	1.000	pass
(l) all combined, 8-bit weights	0.959 ± 0.010	+0.5	0.993	pass
(l) all combined, dt = 0.5 ms	0.823 ± 0.023	-13.1	0.970	FAIL

Pass: |dR@1| <= 2 points and inside the Brian2 seed-to-seed noise band. * Saturated at 100 places: at 2,700 places removing delays costs 10 points (table above). ± is over Poisson seeds.

DESIGN DECISIONS DRIVEN BY THE DATA

Keep synaptic delays, but never store them
no delays: -10 points at 2,700 places; hashed regeneration matches random delays.

Keep the 0.1 ms time step
0.5 ms: -21 points at full scale, runaway activity.

8-bit weights (4-bit validated as fallback)
both inside the noise band; 4-bit halves LSRAM.

1 ms delay bins
no measurable loss; 6 instead of 48 LSRAM blocks per engine.

50 lanes x 2 math blocks per engine
100 lanes would need 800 multipliers (784 exist); 50 lanes use 400 and finish a run in 1.19 ms.

Buffers sized from measured activity
8,452 input spikes per image; 2.41 per step, max 13; <= 57 events per neuron per 1 ms slot.

The fabric is necessary, not optional
Icicle U54 cores: est. 1.9-3.9 s per 4-module query [E]; Jetson Orin Nano GPU, our bit-exact CUDA port: 28 ms per query [M]; fabric target ~1.2 ms [D].

Baselines: DGX Spark and Jetson Orin Nano vs the FPGA Target

Same trained modules, queries and spike-train seeds on every platform | full reports, scripts and raw data in our repository

SAME WORKLOAD ON EVERY PLATFORM: one module run = 784 inputs x 800 neurons x 3,500 steps (dt 0.1 ms)

Platform	Implementation	Per module run	Per 4-module query	Energy/query: above idle / total	Evidence (results/)
DGX Spark: Cortex-X925, 3.9 GHz	Brian2 reference simulator	257 ms [M]	1.03 s [D]	no rail sensors on the DGX	e1_brian2_clean_timing.json
DGX Spark: Cortex-X925, 3.9 GHz	C golden model, 1 core	18.2 ms [M]	72.8 ms [D]	-	e5_bench.json
DGX Spark: Cortex-A725, 2.8 GHz	C golden model, 1 core	32.7 ms [M]	131 ms [D]	-	e5_bench.json
DGX Spark: GB10 GPU	PyTorch float engine (tooling)	169 runs/s [M]	-	-	e5_gpu.json
Jetson Orin Nano: A78AE, 1.73 GHz	Brian2 reference simulator	2.09 s [M]	8.3 s [D]	-	orin_brian2_timing.json
Jetson Orin Nano: 1 A78AE core	C golden model	51.2 ms [M]	205 ms [D]	201 / 1,232 mJ [D]	orin_cpu.json
Jetson Orin Nano: 4 A78AE cores	C golden, 4 modules in parallel	51.2 ms [M]	51.2 ms [M]	152 / 456 mJ [D]	orin_cpu.json
Jetson Orin Nano: GPU, 1.02 GHz	CUDA port (ours, bit-exact)	26.1 ms [M]	28.0 ms [M]	57 / 200 mJ [D]	orin_cuda.json
Jetson Orin Nano: GPU, batched	CUDA port, 96 runs per launch	714 runs/s [M]	-	26 / 55 mJ [D]	orin_cuda.json
Jetson Orin Nano: GPU	PyTorch float engine (tooling)	51.5 runs/s [M]	-	999 / 1,396 mJ [D]	orin_gpu.json
Icicle Kit: U54 RV64GC, 600 MHz	C golden model (290 M instr. [M])	0.48-0.97 s [E]	1.9-3.9 s [E]	to measure (PAC1934)	e5_bench.json
Icicle Kit: FPGA fabric (SpikeNav)	4 engines x 50 lanes, 125 MHz	1.19 ms [D]	~1.2 ms [D]	to measure (PAC1934)	e6_cyclemodel.json

Per 4-module query = 4 x module run when the modules run one after another [D]. Jetson power: INA3221 VDD_IN (whole module), sampled every 10 ms, MAXN_SUPER, idle 5.09-5.11 W; energy above idle = (P_load - P_idle) / rate, total = P_load / rate. Batched energy is per 4 module runs. The DGX has no rail sensors.

Takeaway: per 4-module query, the fabric target (~1.2 ms [D]) is ~23x faster than the best Jetson result (GPU, 28 ms [M]) and ~43x faster than four Jetson CPU cores (51 ms [M]); the Icicle's U54 cores alone would need seconds [E]. Only the fabric is projected to sustain >= 25 queries/s on a 500-place ensemble.

CROSS-PLATFORM CONSISTENCY: bit-identical everywhere [M]

Golden vectors	4 trained modules x 20 queries: identical spike counts on X925, A725, Jetson A78AE, Jetson GPU (CUDA) and an RV64 build under QEMU
Brian2	Jetson reproduces the DGX output spike for spike: 8,206 / 8,206 spikes, E1 test module
Activity traces	1,200 module runs x 3,500 steps of input events: identical; all sizing statistics identical
Ablation re-run	hardware + float configs, 3 model seeds: every threshold and R@1 identical to the DGX
Test suite	13 / 13 tests pass on both machines, incl. the Brian2 exact-raster test

=> the golden vectors are a portable contract: the RTL and the board must match them bit for bit.

WHY AN FPGA: TARGETS SET BY THE JETSON BASELINE

Metric	Jetson Orin Nano (best)	SpikeNav on Icicle (target)
Latency, 4-module query	28.0 ms, GPU [M]	~1.2 ms [D]: ~23x lower
500 places x 3 members (60 module runs/query)	11.9 queries/s, GPU saturated [D]	26-28 queries/s, streaming [D]
Energy, 4-module query	200 mJ total, 57 mJ above idle [D]	< 20 mJ total (O3), PAC1934 on board
Power while running	7.1-9.8 W GPU; 5.1 W idle [M]	< 16.7 W meets O3 [D]
Determinism	bit-exact (our port)	bit-exact vs golden

The Jetson baseline is a strong one, not a straw man
Our CUDA kernel (one block per module run, delay ring in shared memory) is 14x faster than the PyTorch engine and bit-exact. Its limit is structural: 3,500 sequential steps, each with block-wide barriers. The fabric pipelines every step in dedicated lanes, with on-chip weights and delays.

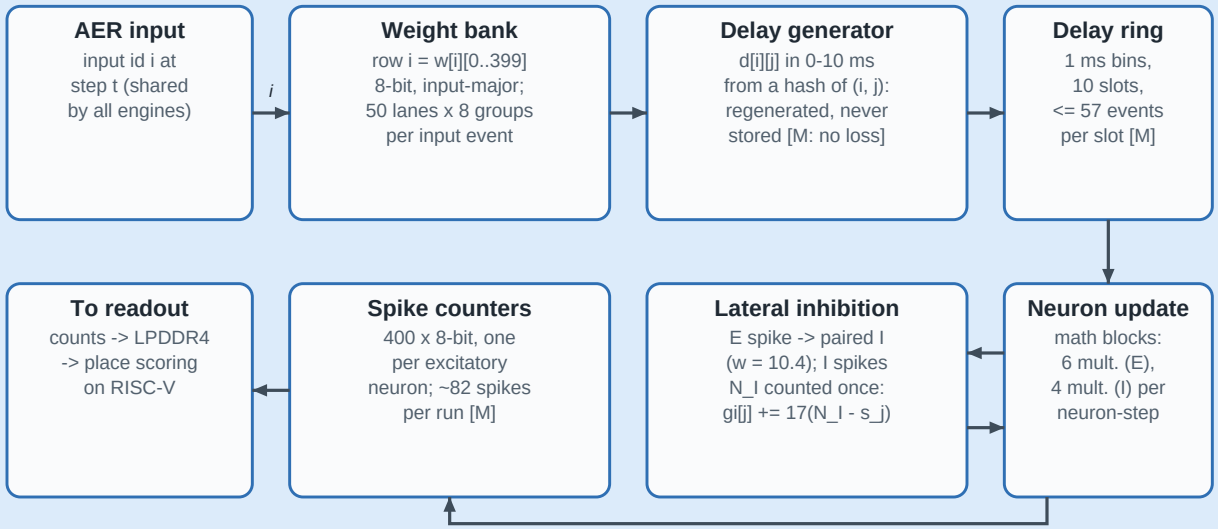
Read the full baseline reports
reports/dgx_spark_baseline_report.md, reports/orin_baseline_report.md and every number's source JSON (results/) in our repository:

github.com/Bejaine/Event-Driven-Spiking-Place-Recognition-for-GPS-Denied-Navigation-on-FPGA

SNN Module Engine: Mapping VPRSNN onto the Fabric

Reference model: Ensemble of Modular SNNs (Hussaini, Milford, Fischer, IEEE T-RO 2024; MIT-licensed code)

ONE MODULE ENGINE (x4). Per 0.1 ms step: accumulate input events, then update 800 neurons



Inhibition needs no 400 x 399 synapse matrix: all I-to-E weights are the constant 17.0, so one shared inhibitory spike count per step replaces ~160k stored synapses (equivalence tested in our suite).

VPRSNN -> HARDWARE MAPPING

Element	Reference model	SpikeNav fabric
Input	28x28 grey, 7x7 patch norm, Poisson px/4 Hz	SmartHLS pre-proc; xorshift spikes [M ok]
Synapses	784 x 400, w in [0,1], random 0-10 ms delay	8-bit LSRAM rows; hashed 1 ms delays
E neurons	conductance LIF, tau 100 ms, theta[j]	integer state on math blocks [M ok]
E<->I	10.4 one-to-one; 17.0 all-but-self	shared I-spike count, O(1) per neuron
Time	dt 0.1 ms, 350 ms per image	3,500 steps; 0.5 ms rejected (-21 pts)
Readout	assignments, hyper-active mask, sum	counts in fabric, scoring on U54
Ensemble, sequence	3-5 members; SeqSLAM, Lseq 2-10	engines in parallel; SeqSLAM on U54

Constants as in the VPRSNN code, including its timer quirk (50 ms effective E lockout), reproduced exactly.

PIPELINE TIMING (cycle model driven by measured event traces)

Stage	Value
Input events: 8,452 per image; 2.41 per 0.1 ms step	measured [M]
Module run, 50 lanes x 2 multipliers, 3,500 steps	148,490 cycles [D]
Module run at 125 MHz (100 MHz)	1.19 ms (1.48 ms) [D]
Option: 100 lanes, constant multiplies in logic	0.62 ms [D]
Front end: DMA + 28x28 pre-processing	< 0.3 ms [E]
Weight swap: 316 KB per module at 1 GB/s	0.32 ms, hidden [D]
Scoring + SeqSLAM + particle filter (U54)	~1-3 ms [E]

Same model in software (per module run)
Brian2 on a 3.9 GHz Cortex-X925 core: 257 ms per image [M].
Our C golden model: 18.2 ms on X925, 32.7 ms on A725 [M].
Jetson Orin Nano: 51.2 ms on one A78AE core, 26.1 ms on its GPU [M].
Icicle U54 at 600 MHz: 290 M RV64 instructions [M] -> 0.48-0.97 s [E].
Neuron updates dominate: input accumulation is the bottleneck on only 3.7% of steps [D], so lanes, not memory, set the speed.

OPERATING MODES (125 MHz, 50 lanes)

Mode	What runs	Expected rate
Resident	4 modules held on-chip (100 places, the paper's robot-trial size)	~1.2 ms per query [D] (+ < 0.3 ms front end)
Streaming	all modules of all members; 2 engines compute, 2 banks load	500 places x 3 members: 28 queries/s at >= 1 GB/s; 26 at 0.5 GB/s [D]
Prior-gated	only modules near the localiser's belief	~1.2 ms, independent of route length [D]
Batch benchmark	each module loaded once, all queries run through it	full Nordland (3,300 x 132 runs) ~2-3 min [D]

Resource Budget, Hardware / Software / IP, and Verification Flow

Identification of required hardware, software, interfaces and FPGA IP cores | baseline: 4 engines x 50 lanes at 125 MHz

FPGA RESOURCE BUDGET (MPFS250T)

Block	Logic (LE)	LSRAM	Math blocks
4 weight banks (8-bit, x40)	-	520 [D]	-
4 delay rings (1 ms bins)	-	24 [D]	-
4 engine datapaths (50 x 2 mult.)	~40-55K [E]	~8 [E]	400 [D]
Pre-processing + spike generators	~10-15K [E]	~10 [E]	~16 [E]
DMA, interconnect, CSRs, readout	~15-20K [E]	~20 [E]	0
Total	~65-90K	~582	~416
Device capacity	254K	812	784
Utilisation	~26-35%	~72%	~53%

How the numbers were obtained

LSRAM and math-block counts come from our cycle model [D]; logic is estimated [E] and will be confirmed by Libero synthesis in the first weeks.
Module = 784 x 400 weights + theta + labels + mask = 316,054 bytes [D].
LSRAM is the binding resource, so:

- resident mode holds 4 modules (100 places) entirely on-chip;
- larger maps stream modules from the LPDDR4 library through the same banks;
- 4-bit weights (validated within noise [M]) need 260 instead of 520 blocks.

Delays are hash-regenerated, and inhibitory synapses are not stored at all.
100 lanes per engine (0.62 ms) needs 800 multipliers > 784: possible only with constant-coefficient decays moved to shift-add logic (an optimisation, not baseline).

Module library and training cost

100 places = 4 modules = 1.26 MB; 500 places x 3 members = 60 modules = 19 MB;
a 1.5 GB share of the LPDDR4 holds ~4,700 modules.
Training 500 places x 3 members with the reference code: ~31 core-hours,
~3.1 h on the 18-core workstation we used [D].

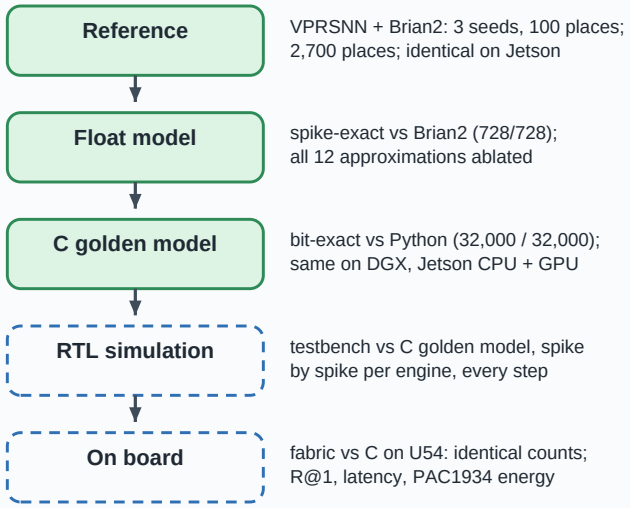
Clocking and interfaces

fabric 125 MHz (PF_CCC); MSS <-> fabric over FIC AXI4;
LPDDR4 via the MSS DDR controller; PAC1934 and IMU over I2C.

HARDWARE, SOFTWARE AND IP CORES

Category	Items
Board	PolarFire SoC Icicle Kit (MPFS250T, 2 GB LPDDR4, USB OTG, GbE, PAC1934, mikroBUS, RPi header)
Team-supplied	USB UVC webcam; IMU breakout; microSD; 12 V battery pack; handheld rig or small rover
Microchip IP	PolarFire SoC MSS, FIC0/FIC1 AXI4, CoreAXI4Interconnect, CoreAXI4DMAController, PF_TPSRAM / PF_DPSRAM (LSRAM), PF_URAM (uSRAM), PF_CCC, Math block (MACC)
Our RTL / HLS	pre-processing, spike generator, AER broadcast, module engines, weight streamer, readout
Built already	float + fixed-point Python models, bit-exact C golden model, cycle model, 13-test suite; DGX Spark + Jetson baselines; bit-exact CUDA port
Tools	Libero SoC + SmartHLS, SoftConsole, Yocto Linux (Microchip BSP), simulator testbenches
Software	VPRSNN (MIT) + Brian2 reference; Linux drivers, scheduler, scoring, SeqSLAM, filter

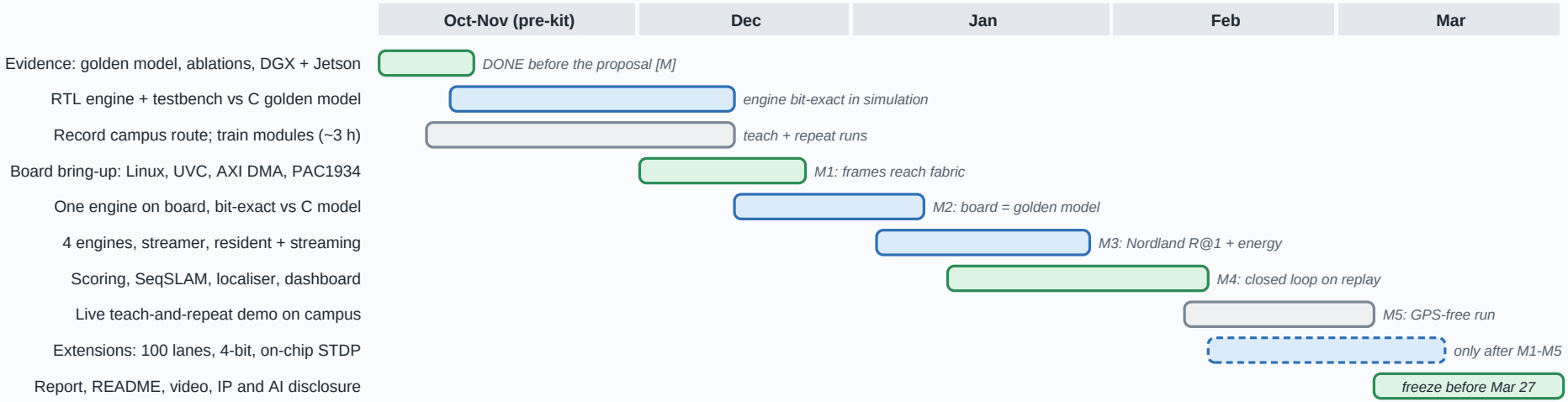
VERIFICATION FLOW (solid = done, dashed = planned)



Objectives, Development Plan, and Risk Mitigation

Design and Development phase: Nov 30, 2026 - Mar 27, 2027 (17 weeks); evidence and baselines already complete

SCHEDULE AND MILESTONES



Roles: (1) SNN reference, training, datasets, golden model; (2) RTL / SmartHLS engines and verification; (3) Linux, drivers, scoring, localiser, dashboard and demo platform. Pre-kit RTL simulation absorbs shipping delays.

OBJECTIVES AND MEASURABLE SUCCESS CRITERIA

#	Objective	Success criterion / status
O1	Faithful VPRSNN inference in fabric: board counts bit-identical to the C golden model	R@1 within 2 pts of float; already -1 to -2 pts at 2,700 places in simulation [M]
O2	Real-time inference, including ensembles	~1.2 ms per query, ~23x below Jetson GPU 28 ms [M]; >= 25 q/s for 500 places x 3 members [D]
O3	Energy efficiency, measured on the board (PAC1934)	< 20 mJ per query: >= 10x below Jetson GPU 200 mJ [D] and C on the U54 cores
O4	GPS-free teach-and-repeat on a campus route	bounded route error vs odometry-only drift
O5	Open-source engine, golden and cycle models, benchmarks	public repo (DGX + Jetson baselines already online); tests + README reproduce

Attribution and disclosure

Builds on VPRSNN (Hussaini, Milford, Fischer, IEEE T-RO 2024, MIT License), based on Diehl and Cook (2015); SeqSLAM (Milford and Wyeth, 2012). Hardware architecture, fixed-point, golden, cycle and CUDA models: Merconez. AI disclosure: Claude (Anthropic) assisted drafting; Claude Code implemented evaluation, baseline and CUDA scripts under team direction. All results are reviewed by the team and reproducible from our repository:

github.com/Bejaine/Event-Driven-Spiking-Place-Recognition-for-GPS-Denied-Navigation-on-FPGA

KEY RISKS AND MITIGATIONS

Risk	Mitigation / evidence
Approximations hurt accuracy	retired pre-proposal: combined configuration -1 to -2 R@1 points at 2,700 places [M]
Math-block budget	baseline 50 lanes uses 400 of 784 [D]; 100 lanes only with constant multiplies in logic
Time step too coarse	0.5 ms rejected (-21 points) [M]; fixed at 0.1 ms
LSRAM capacity	~582 of 812 at 8-bit [D]; 4-bit fallback [M ok]
DDR bandwidth	compute-bound from 0.5 GB/s at 50 lanes [D]; measure FIC throughput at M1
Timing closure	pipelined lanes; at 100 MHz a run takes 1.48 ms [D]
Viewpoint sensitivity (stated in the paper)	same-path teach-and-repeat, fixed camera mount, sequence matching
Kit arrives late	RTL verified against the C golden model pre-kit
An edge GPU is enough	measured, not assumed: Jetson Orin Nano GPU 28 ms, 200 mJ per query [M/D]; target 1.2 ms
Scope creep	extensions only after milestone M5